

Quantifying forest loss from wildfires

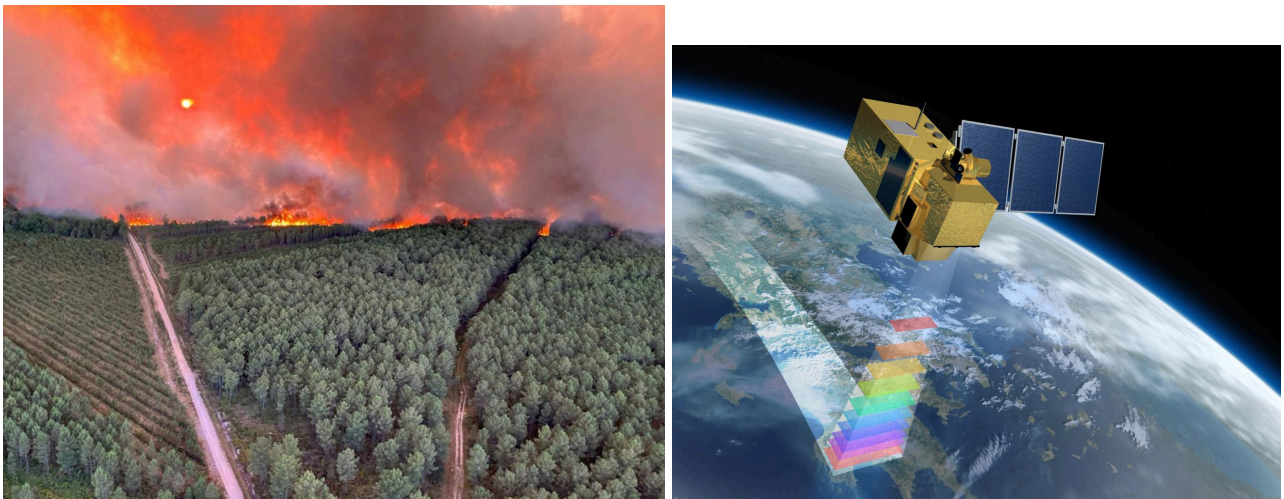
Case study: La Teste-de-Buch (Gironde, France) fire in July 2022

Master Smart Data ENSAI 2026-2027

Machine learning for Earth observation and remote sensing

Supervisor: Florian Lalande

This is a group project for teams of 2 to 3 students. Choose your teammates carefully: you will be spending some quality time together, staring at satellite pixels.



[left] Wildfire in La Teste-de-Buch in July 2022 (Credits: Sdis33/Service communication for SudOuest) ; [right] Rendering of the Sentinel-2 satellite in orbit (Credits: Copernicus program, ESA)

This project relies on Python only. Jupyter notebooks can be used. Everything you need (satellite imagery, data labels, land-cover maps, etc) can be found on Google Earth Engine, which is free of use for research and education purposes. The project is divided into 5 steps.

One thing before you start, and please read it twice: I care far more about how you think than about your final results. More on that at the very end.

Step 0: Understand what happened

In July 2022, the South West of France caught fire. The department of Gironde burned through more than 20,000 hectares during that month, and roughly 40,000 people were evacuated. One of the two major wildfires started near La Teste-de-Buch on 12 July, right next to the Dune du Pilat. By the time it was contained, somewhere between 6,000 to 7,000 hectares of pine forest was burnt.

Before you touch a single pixel, please spend some time understanding what actually happened. There is a Wikipedia page on the 2022 Gironde fires, called “Incendies de 2022 dans la forêt des Landes”. It is only available in French as far as I know, so feel free to use automatic translation if you need to.

This was not a single one-off wildfire. As heatwaves are getting stronger and more frequent, droughts longer, and forests drier, fires like this one will become more common. Monitoring them from space is not just an academic exercise, but it is how we keep monitoring a situation that is going to become more and more problematic. And this is the whole point of this project.

Deliverable for this step:

- In your own words, please write a small introduction for this project, summarizing what happened at La Teste-de-Buch in July 2022 and why remote sensing of wildfires matters.

Step 1: Setting up Google Earth Engine

Google Earth Engine (GEE) is free for research and education. Signing up takes a few minutes. Please follow these steps (or find another tutorial online if you are stuck):

- Go to earthengine.google.com and register with a Google account.
- Request access for **non-commercial / research use** (this is the free tier).
- Create a Cloud project when prompted. Approval is usually quick.

A word on the data: Sentinel-2

You will be using Sentinel-2, part of the European Copernicus program. This is a pair (actually, now a trio) of satellites carrying a multispectral camera with 13 spectral bands. The Sentinel-2 satellites revisit the same spot every 5 days, and its sharpest bands see the ground at 10 meters per pixel.

In this project, we will use the Level-2A product (surface reflectance). The name of the collection on GEE is `COPERNICUS/S2_SR_HARMONIZED`. "Surface reflectance" means the effect of the atmosphere has mostly been mitigated, so you are looking at the ground rather than the haze above it. These images are ready for machine learning use.

A note on the band names: we usually write bands as B2, B3, B4, and so on. That is how GEE labels them. Please have a look at the Sentinel-2 dataset catalog below.

Further reading:

- [\(ESA\) Sentinel-2 mission overview](#)
- [\(Copernicus\) Sentinel-2 data documentation](#)
- [\(Google Earth Engine\) Sentinel-2 dataset catalog](#) ← use surface reflectance

No deliverable for this step.

Step 2: Visualize the fire

Now, we start with the first real task. You will visualize the fire.

Using Python scripts or Jupyter Notebooks, download individual Sentinel-2 images over La Teste-de-Buch. Download all available images for the month of July 2022. I provide the exact area of interest (AOI) in the following table (bounding box). Coordinates are in longitude/latitude (called EPSG:4326).

Coordinate	Value (degrees)
Longitude min	-1.23
Longitude max	-0.89
Latitude min	44.72
Latitude max	44.96

Reproject the pixels to EPSG:32630 (UTM zone 30N) so your pixels are measured in actual meters, and not degrees. Degrees are fine until you try to count real distances or areas, so we are going to use EPSG:32630. You can just specify the export coordinates into GEE, no need to derive the reprojection by hand.

For now, just download the three visible bands: B2 (blue), B3 (green), B4 (red). We will make normal-looking color images. Make the RGB “true color” images from these three visible bands.

Then build an animation from the July images and describe what you see. Where is the burnt forest scar? When does it appear? How fast does it grow?

Deliverables for this step:

- Show the animation in RGB (true color) for all available Sentinel-2 images in July 2022 over the specific area of interest in La Teste-De-Buch.
- Describe the generation process and comment on what is happening in the animation.

Step 3: Making yearly composites (i.e. getting rid of clouds)

Single images are pretty, but they are flawed. In particular, clouds and shadows get in the way, and individual images cannot be fairly compared between each other. To compare them, we will need something cleaner.

The trick is to build yearly composite images, also called basemaps. It works as follows: you take multiple images over a fixed period (in our case, one year), and for each pixel you keep the median value across all images. Noise and seasonal variations will be averaged out, and this will result in a stable image for the whole period. In addition, we will also filter out clouds before taking the median.

Build one composite image per year for 2020, 2021, 2022, 2023, 2024. Use six bands this time: B2 (blue), B3 (green), B4 (red), B8 (NIR), B11 (SWIR1) , B12 (SWIR2). These extra three bands will be useful to investigate vegetation health and burn scars.

As mentioned above, we will filter out clouds before computing median composite. For this, use the S2 cloud probability layer, `COPERNICUS/S2_CLOUD_PROBABILITY`, to filter out cloudy pixels before taking the median. The S2 Cloud Probability layer gives each pixel a score between 0 and 100, indicating the likelihood for this pixel being a cloud.

You will have to decide how to remove the clouds by choosing a threshold. If too strict, you delete a lot of good data. But if too loose, clouds can remain into the composite image. Try several values, and show results for a few of them. Then make a choice and explain your reasoning.

Finally, make another RGB (in true colors) animation of the five yearly composite images and comment on it. What changes between 2021 and 2022? Between 2022 and 2023?

Deliverables for this step:

- Visualize side by side an image from Sentinel-2 in RGB (true color) and its corresponding S2 Cloud Probability layer. Comment on the probability values.
- Try multiple cloud filtering thresholds. Show a few yearly composite images at various thresholds, comment and justify your final threshold choice.
- Make the final yearly composite images (with all six bands), and make a small animation in true colors for the final result. Comment on the final result.

- **[BONUS]** On top of true color images, generate false color images using the common [B8, B4, B3] false color coding. Discuss the results and explain what the colors now represent.

Step 4: The main event: semi-supervised learning

The real data science and machine learning task starts from here.

(A) Download pseudo-labels: ESA WorldCover

To train a classifier we will need labels, i.e. pixels someone has already sorted into categories. You will borrow them from the ESA WorldCover, a global 10 m land-cover map. This dataset is available on GEE for the year 2021, and the collection name is "ESA/WorldCover/v200". Download the layer for the area of interest.

These are not ground truth handed down from the heavens. They are a model's best guess. This is why we refer to them as “pseudo-labels”, as their reliability is not perfectly granted.

Deliverables for this step:

- Visualize the ESA WorldCover labels side by side with your 2021 basemap.
- Look hard at the pseudo-labels, and make some exploratory statistics. Comment extensively, there should be a lot to say.

(B) Train two classifiers

Now, we will start the modeling part. You will train two classifiers on the WorldCover labels: one Random Forest and one Convolutional Neural Network (CNN).

First, you will compute some features (also called “spectral indices” in remote sensing). Raw bands are useful, but certain band combinations are far more insightful. Compute these four spectral indices per pixel:

Index	Formula
NDVI	$(B8 - B4) / (B8 + B4)$
NDWI	$(B3 - B8) / (B3 + B8)$
NDBI	$(B11 - B8) / (B11 + B8)$
BSI	$((B11 + B4) - (B8 + B2)) / ((B11 + B4) + (B8 + B2))$

So now, each pixel ends up with 6 bands + 4 indices = 10 features for each year.

Now, train both models using the pseudo-labels. Assess the performances of each model properly. Split your data, report accuracy and a confusion matrix. Next, apply both trained models to your other yearly basemaps (2020, 2022, 2023, 2024). Save your outputs (i.e. the classified maps) as you will need them for the last step.

A reminder: your labels are imperfect, so perfect accuracy would in fact be suspicious. A model that matches noisy labels 100% has usually just memorized the noise.

Deliverables for this step:

- Make some research on the four spectral indices provided in this section. Explain what the acronyms mean and what each index is supposed to highlight. Show some maps in your report, and comment.
- Present your methodology and extensively document the model training and evaluation phases. Show plots when appropriate. This is essentially the biggest chunk of this project, where you can shine by demonstrating the methods and rigour you learned at ENSAI. Don't be afraid to do too much.

(C) The final map!

We are almost there. Using your yearly classifications, make one final map of vegetation lost during the 2022 La Teste-de-Buch fire.

Then, derive the total area of burnt forest. If you are still working with the EPSG:32630 coordinate system (which you should), each pixel is $10 \times 10 \text{ m} = 100 \text{ m}^2$. So converting pixels into hectares is straightforward. Report how many hectares of vegetation were lost, and compare your number to the reported data on Wikipedia (or any other online source).

If you are wildly off, that's completely fine, do not try to hide it. Instead, explain why (so many things could have gone wrong in the process until here). I genuinely do not care and would prefer a wrong number that you understand over a correct number you cannot defend.

Everything here should be visual and every choice documented.

Deliverables for this step:

- Do what is above

How I will grade the project

I care about how you think, and not whether your final hectare calculation matches mine down to the last digit. A beautiful map with no reasoning behind it will score lower than a rough one where every decision is properly justified.

Specifically, I am looking for four things:

- **Thorough understanding** → What happened on the ground? What data are you playing with? Please show me that you thoroughly understand what we are trying to achieve.
- **Reasoning** → How do your steps follow from one another?
- **Justification** → Why did you choose a specific method? Can you justify the parameters you selected (e.g. the cloud probability threshold, the CNN hyper-parameters, etc).
- **Critical thinking** → Did you question the data? The pseudo-labels? Your own results? Can you notice when something looks too good or too wrong?

The final result matters, but it matters less than the road you took to get there. Show me you understand what you did. That is the whole point of this project.

Also, I purposefully kept the instruction pretty vague for the last part of this project (step 4). Make reasonable assumptions, document and justify them in your report.

Do not hesitate to reach out to me via e-mail, I will do my best to reply asap. Please do not wait for the last moment to tell me that you did not understand the initial steps. Seek help early if needed, it is totally fine.

Good luck!